

TEMPERATURE MONITORING On Cayenne IoT Platform

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This temperature monitoring system is based on Cayenne Internet of Things (IoT) platform, NodeMCU microcontroller (MCU) and Message Queuing Telemetry Transport (MQTT) protocol. The authors' prototype is shown in Fig. 1.

Cayenne accelerates the development of IoT-based solutions, including quick design, prototyping and other commercialised projects. It is a drag-and-drop IoT project builder that can help developers build complete, ready-to-use IoT solutions with little to no coding.

Cayenne contains a vast catalogue of certified IoT-ready devices and con-

nectivity options. This allows users to easily add any device to the library utilising MQTT API. All devices in Cayenne are interoperable and benefit from features such as rules engine, asset tracking, remote monitoring and control, and tools to visualise real-time and historical data.

MQTT is a lightweight messaging protocol for sensors and mobile devices.

Circuit and working

The circuit diagram of the temperature monitoring system is shown in Fig. 2. Hardware used in the project include NodeMCU (ESP8266-12E), temperature sensor (LM35) and 5V power supply.

NodeMCU. NodeMCU ESP8266-12E MCU is a development board with one analogue and many general-purpose input output (GPIO) pins. It has 4MB flash memory, and can operate at a default clock frequency of 80MHz. In this project, analogue pin A0 of

NodeMCU is used to read output voltage of LM35 temperature sensor.

LM35. This is a precision IC temperature sensor. It provides output voltage that is linearly-proportional to temperature (in °C). It exhibits typical accuracies of $\pm 1/4^{\circ}\text{C}$ at room temperature and $\pm 3/4^{\circ}\text{C}$ over a full temperature range from -55 to $+150^{\circ}\text{C}$.

LM35 provides sensitivity of $10\text{mV}/^{\circ}\text{C}$. In this project, Vout pin (pin no. 2) of LM35 is connected to pin A0 of NodeMCU. Pins 1 and 3 of LM35 are connected to +5V and Gnd, respectively.

NodeMCU programming with Arduino IDE

You need Arduino IDE software to program NodeMCU. Download the latest Arduino IDE from www.arduino.cc/en/Main/Software and install it on your computer.

Steps for adding NodeMCU ESP8266 board and library to Arduino IDE are given below.

1. Open Arduino IDE. Go to File menu and select Preferences.

2. Copy http://arduino.esp8266.com/stable/package_esp8266com_index.json and paste in Additional Boards Manager URL.

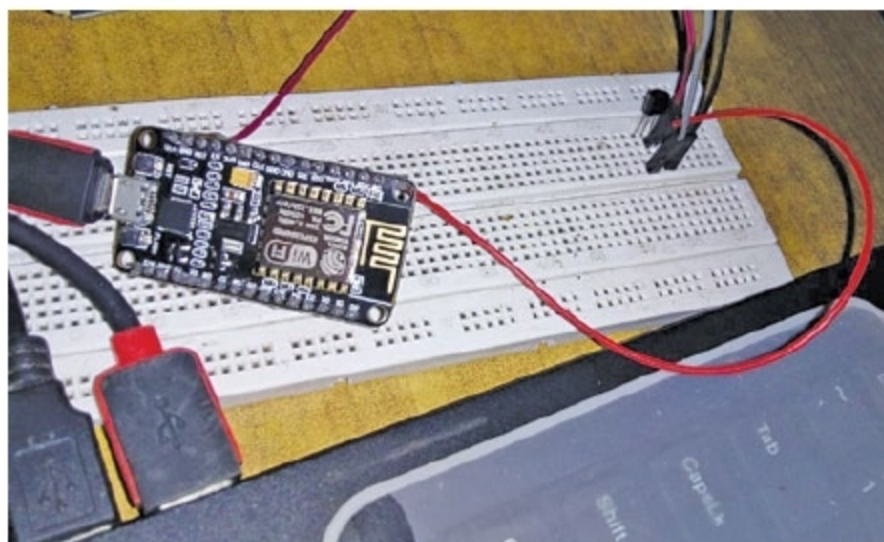


Fig. 1: Authors' prototype

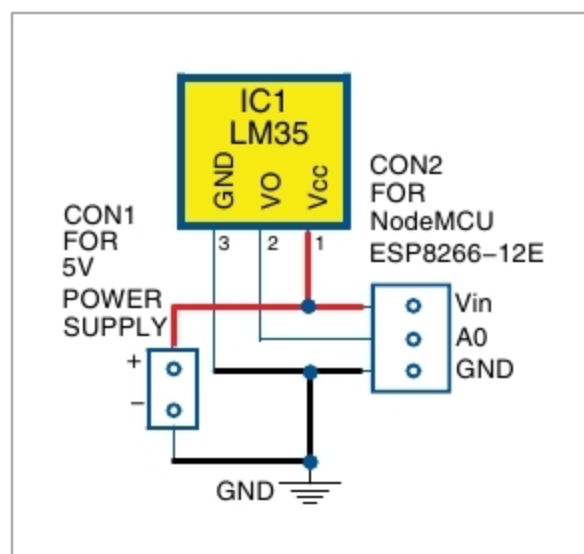


Fig. 2: Circuit diagram of temperature monitoring system

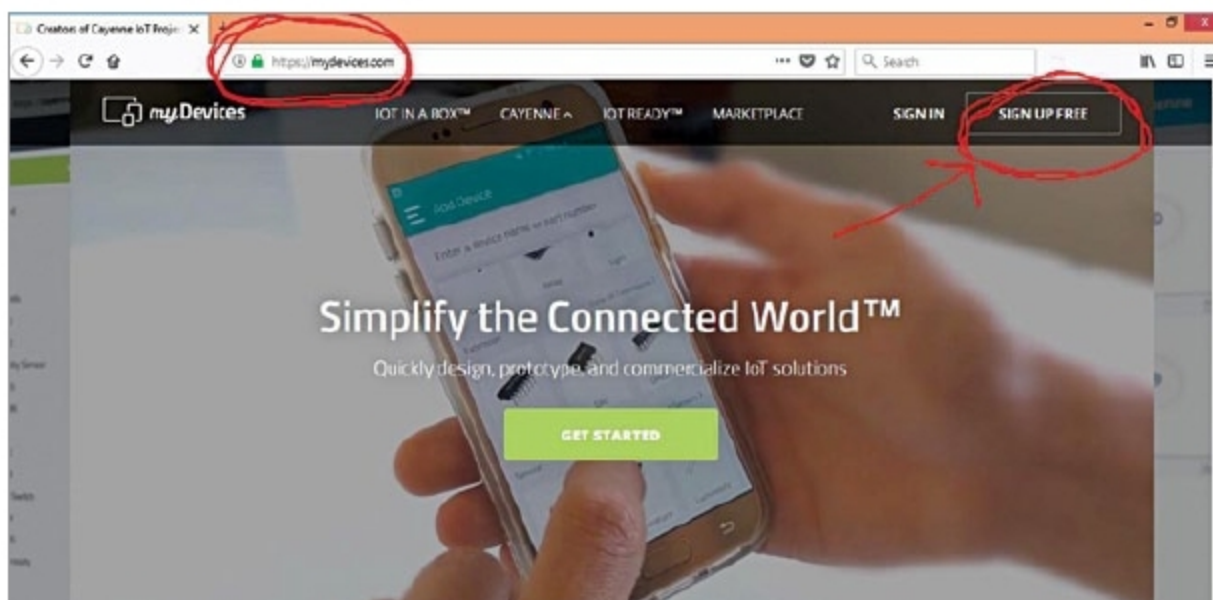


Fig. 3: Cayenne sign up